



PROJECT REPORT

Software Engineering (CSE327)

Project Title: AI-Inspired E-Commerce Platform
(Projukti Mart - AI-Inspired Electronics Web Marketplace)

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1. Introduction

1.1 || Purpose

This Software Requirements Specification (SRS) document provides a complete description of the functional and non-functional requirements for **Projukti Mart**, an AI-inspired e-commerce platform specialized in electronic devices. The document is intended for the development team, project stakeholders, instructors, and future maintainers. It serves as the primary reference for system modeling, design, implementation of the prototype, and validation activities required by CSE327.

1.2 || Scope

Projukti Mart is a multi-sided online marketplace that enables customers to discover, evaluate, and purchase electronic products (Laptops, Desktop PCs, Keyboards, Headphones, Cables, and Stands), allows sellers to manage product listings and fulfill orders, and provides platform administrators with oversight and policy-control capabilities. The system embeds a mandatory **AI-Assisted Semantic Product Search** capability that accepts natural-language queries and returns ranked, relevant products. On product listing and detail pages, the system also surfaces light, personalized recommendations based on the user's recent browsing and purchase history. When the AI component is unavailable, the platform degrades gracefully to conventional keyword/filter search.

In scope:

- User registration, authentication, and profile management.
- Product catalog browsing, filtering, and AI-assisted search.
- Shopping cart, checkout, order placement, and order history.
- Product ratings and reviews.
- Seller product listing management and order processing.
- Basic sales summaries for sellers.
- Administrator management of categories, sellers, users, and flagged content.
- AI semantic search + personalized recommendations with fallback behavior.

Out of scope (for this project):

- Payment gateway integration with real financial institutions (mock payment is acceptable).
- Physical logistics/shipping carrier integration.
- Mobile native applications (responsive web is sufficient).
- Multi-language support beyond English.

1.3 || Definitions, Acronyms, and Abbreviations

Term	Definition
AIEP	AI-Inspired E-Commerce Platform (project context)
SRS	Software Requirements Specification
AI Component	The semantic search/recommendation service is integrated into Projukti Mart
Semantic Search	Natural-language query understanding that matches intent and product attributes beyond exact keywords
Fallback Search	Conventional keyword + filter search is used when the AI component is unavailable or returns low-confidence results
Customer	Registered or guest user who browses and purchases products
Seller / Vendor	Registered business or individual who lists and sells products
Admin	Platform Administrator with elevated privileges
SKU	Stock Keeping Unit – unique identifier for a sellable product variant
Line Item	A single product entry within an order, including quantity and unit price
NFR	Non-Functional Requirement

1.4 || Overview

The remainder of this document is organized as follows. Section 2 presents the overall product perspective, high-level functions, user characteristics, constraints, and assumptions. Section 3 details the functional and non-functional requirements, data descriptions, interfaces, system rules, and role permissions. Section 4 describes external interfaces, including the AI feature interface, which contains all required modeling and design artifacts: use-case diagrams and descriptions, sequence diagrams, activity diagrams, class diagrams, architecture diagrams, and conceptual UI wireframes.

2. Overall Description

2.1 || Product Perspective

Projukti Mart is a new, self-contained web-based e-commerce system. It is not an extension of an existing product. The platform follows a classic three-tier architecture (presentation, application/business logic, data) with an additional loosely coupled AI service layer. The AI service may be implemented as a lightweight local model, rule-based semantic matching, or a call to an external LLM/embedding API; the rest of the system treats it as a replaceable component behind a well-defined interface. The system interacts with end users through a responsive web UI. Internal components communicate via REST-style service calls or direct method invocations within the same process (prototype decision).

2.2 || Product Functions (High-Level)

Customer-facing functions

- Register, log in, and manage personal profile and addresses
- Browse product catalog by category and apply filters (brand, price, specs)
- Perform AI-assisted natural-language product search
- View personalized product recommendations
- Add items to cart, update quantities, and place orders
- Track order status and view order history

- Rate and review purchased products

Seller-facing functions:

- Create, update, and deactivate product listings (including stock and price)
- View and process incoming orders for own products
- View basic sales summaries (units sold, revenue) for own catalog

Administrator functions:

- Manage product categories and sub-categories
- Approve/suspend sellers and manage user accounts and roles
- Monitor orders, flagged reviews, and suspicious activity
- Define platform-wide catalog policies (e.g., allowed categories)

AI-driven functions:

- Accept natural-language search queries and return ranked product results
- Generate personalized “You may also like” recommendations
- Gracefully fall back to keyword search when AI confidence is low, or the service is down

2.3 || User Characteristics

Primary users are expected to be comfortable with standard web browsers and online shopping flows. Technical sophistication varies:

Role	Characteristic
Customer	General consumers or students/professionals seeking electronics. May issue natural-language queries rather than structured filters. Expect fast search and clear product specifications.
Seller	Small-to-medium electronics retailers or individual resellers. Need simple inventory and order management tools—moderate technical skill.
Administrator	Platform operators with higher technical literacy. Responsible for policy enforcement, content moderation, and user management.

2.4 || Constraints

Catalog Structure:

- A product hierarchy exists: Category → Sub-category → Product.
- Each product belongs to exactly one seller and one leaf category.

Ordering Constraints:

- An order cannot be placed for a quantity exceeding available stock.
- Each order belongs to exactly one customer and may contain multiple line items.
- Order total equals the sum of line-item subtotals + applicable shipping/tax (rules defined by the team).

AI Feature Constraints:

- The AI feature must be functionally demonstrable (rule-based, lightweight model, or external API).
- The system must degrade gracefully to keyword search if the AI component is unavailable.
- Any third-party AI API usage must be documented (role, inputs, outputs). Technical / Project Constraints
 - Prototype focus is on core workflows and AI integration, not production-grade scaling or security hardening.
 - Web-based interface is mandatory; native mobile apps are out of scope.

2.5 || Assumptions and Dependencies

- Users have access to a modern web browser (Chrome, Firefox, Edge, or Safari) and an internet connection.
- Product data (titles, descriptions, key specifications) is sufficiently rich for semantic matching.
- If an external LLM/embedding API is used, network access and valid API credentials are available at the time of the demonstration.
- Mock payment is acceptable; real payment gateway integration is not required.
- Sellers are responsible for the accuracy of the stock levels they maintain.

- The development team will provide a representative sample of data covering all six product families.
- English is the sole supported language for UI and natural-language queries in the prototype.

3. System Requirements

3.1 || Functional Requirements

Requirements are identified with the prefix FR- followed by a unique number.

Authentication & Profile:

ID	Requirement Statement
FR-01	The system shall allow a new user to register using an email address, password, name, and role (Customer or Seller).
FR-02	The system shall authenticate users via email and password and maintain a session.
FR-03	The system shall allow a logged-in user to view and update profile information and shipping addresses.
FR-04	The system shall support role-based access control (Customer, Seller, Administrator).

Catalog & Search:

ID	Requirement Statement
FR-05	The system shall present a hierarchical catalog: Category → Sub-category → Product.
FR-06	The system shall allow customers to browse products by category and apply filters (price range, brand, key attributes).
FR-07	The system shall accept natural-language search queries and return a ranked list of relevant products via the AI component.
FR-08	When the AI component is unavailable or returns low confidence, the system shall fall back to keyword/filter search and inform the user.
FR-09	The system shall display product detail pages that include the title, description, specifications, price, stock status, seller, images, and reviews.

FR-10	On product listing and detail pages, the system shall display personalized recommendations based on the user's recent browsing/purchase history.
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Cart & Checkout:

ID	Requirement Statement
FR-11	The system shall allow a customer to add products to a shopping cart with a chosen quantity.
FR-12	The system shall prevent adding a quantity that exceeds the current available stock.
FR-13	The system shall allow the customer to update quantities or remove items from the cart.
FR-14	The system shall calculate the order total as the sum of line-item subtotals plus any shipping/tax defined by the team.
FR-15	The system shall allow a logged-in customer to place an order from the cart contents
FR-16	Upon successful order placement, the system shall decrement stock for each line item and record the order.

Orders & Reviews:

ID	Requirement Statement
FR-17	The system shall allow a customer to view order history and current order status.
FR-18	The system shall allow a customer to cancel an order only while it is in an allowed state (e.g., Pending).
FR-19	The system shall allow a customer who has purchased a product to submit a rating (1-5) and a textual review.
FR-20	The system shall display product reviews on the product detail page.

Seller Operations:

ID	Requirement Statement
FR-21	The system shall allow a seller to create, update, and deactivate product listings (title, description, specs, price, stock, category, images).
FR-22	The system shall allow a seller to view and update the status of orders containing their products.
FR-23	The system shall provide a seller with a basic sales summary (units sold and revenue) for their products.

Administration:

ID	Requirement Statement
FR-24	The system shall allow an administrator to create, edit, and deactivate categories and sub-categories.
FR-25	The system shall allow an administrator to manage user accounts (activate, suspend) and assign roles.
FR-26	The system shall allow an administrator to view platform-wide orders and flag or moderate reviews.
FR-27	The system shall allow an administrator to suspend a seller account.

AI Component Behavior:

ID	Requirement Statement
FR-28	The AI component shall accept a natural-language query string and optional context (user ID, category filter).
FR-29	The AI component shall return a ranked list of product identifiers with relevance scores.
FR-30	If the AI component cannot produce a confident result, it shall signal low confidence so the system can invoke fallback search.
FR-31	The recommendation service shall accept a user identifier (or anonymous session) and

	return a list of recommended product identifiers.
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3.2 || Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Keyword search results shall be returned in under 2 seconds for catalogs up to 5,000 products (prototype scale).
NFR-02	Performance	AI semantic search shall return results in under 5 seconds under normal load, or the system shall time out and fall back.
NFR-03	Reliability	If the AI service is unreachable, the platform shall remain fully usable via keyword search and display a non-blocking notice.
NFR-04	Usability	Primary customer workflows (search → product → cart → checkout) shall be completable in fewer than 8 clicks/taps from the homepage.
NFR-05	Security	Passwords shall be stored using a one-way hashing algorithm. Sessions shall expire after a period of inactivity.
NFR-06	Security	A seller shall only be able to modify products and view orders that belong to them.
NFR-07	Scalability	The architecture shall allow the AI component to be replaced or scaled independently of the core application.
NFR-08	Maintainability	Source code and documentation shall follow consistent naming and structure so that a new team member can locate major modules within 30 minutes.
NFR-09	Availability	Core catalog, cart, and order functions shall be available even when the AI service is down.
NFR-10	Compatibility	The web UI shall be usable on the latest two major versions of Chrome, Firefox, Edge, and Safari at desktop resolutions.

3.3 || Data Descriptions

Key entities and their principal attributes:

Entity	Key Attributes
User	userId, email, passwordHash, fullName, role {Customer Seller Admin}, status, createdAt
Address	addressId, userId, line1, line2, city, postalCode, country, isDefault
Category	categoryId, name, parentCategoryId, description, isActive
Product	productId, sellerId, categoryId, title, description, brand, price, stockQty, specs, images, status, createdAt, updatedAt
Cart	cartId, userId (or sessionId), items[] {productId, quantity, unitPrice}
Order	orderId, customerId, status {Pending Confirmed Shipped Delivered Cancelled}, totalAmount, shippingAddress, createdAt, updatedAt
OrderItem	orderItemId, orderId, productId, quantity, unitPrice, subtotal
Review	reviewId, productId, customerId, rating (1–5), comment, createdAt, isFlagged
SearchQueryLog	queryId, userId (nullable), rawQuery, usedAI (bool), resultCount, timestamp

3.4 || Interface Descriptions

The primary user interface is a responsive web application. Key interaction points with the AI component are:

- (1) the search bar that accepts free-text natural-language queries,
- (2) the search results page that displays ranked products and keyword mode was used
- (3) Recommendation carousels on the home page and product detail pages. Internal service interfaces are described in the Architecture section.

3.5 || System Rules and Platform Constraints

- Stock cannot become negative; order placement must atomically check and decrement stock.
- A product may belong to only one seller and one leaf category.
- Only the owning customer or an administrator may cancel an order, and only while the order is in an allowed state (Pending / Confirmed).
- A customer may review a product only after at least one order for that product has been delivered.
- Sellers can edit only their own product listings.
- Administrators may suspend users and flag reviews; flagged reviews are hidden from public view until moderated.

3.6 || User Role Permissions

Action	Customer	Seller	Admin
Register / Login	Yes	Yes	Yes
Browse & Search catalog	Yes	Yes	Yes
Use AI semantic search	Yes	Yes	Yes
Add to cart / Place order	Yes	No*	No*
View own orders	Yes	Yes (as seller)	Yes (all)
Write review	Yes (purchased)	No	No
Manage own products	No	Yes	Yes (override)
Process own orders	No	Yes	Yes
Manage categories	No	No	Yes
Manage users/roles	No	No	Yes
Flag / moderate reviews	No	No	Yes

* Sellers and Admins may also act as customers if they hold a dual role; otherwise, purchase is restricted to the Customer role.

4(a). External Interface Descriptions

4(a).1 || User Interface Expectations

The UI shall present a clean, modern electronics store aesthetic with a persistent header containing a logo, category navigation, a cart icon, and a user menu. Product cards shall show image, title, key specs, price, and rating. Recommendation sections shall be labeled “Recommended for you” or “Customers also viewed”.

4(a).2 || AI Feature Interface

Semantic Search Service:

- **Input:** natural-language query string, optional userId, optional category filter, max results.
- **Output:** ordered list of {productId, relevanceScore, explanationSnippet (optional)}; confidence flag.
- **Failure mode:** timeout or explicit low-confidence response → caller invokes keyword search.

Recommendation Service:

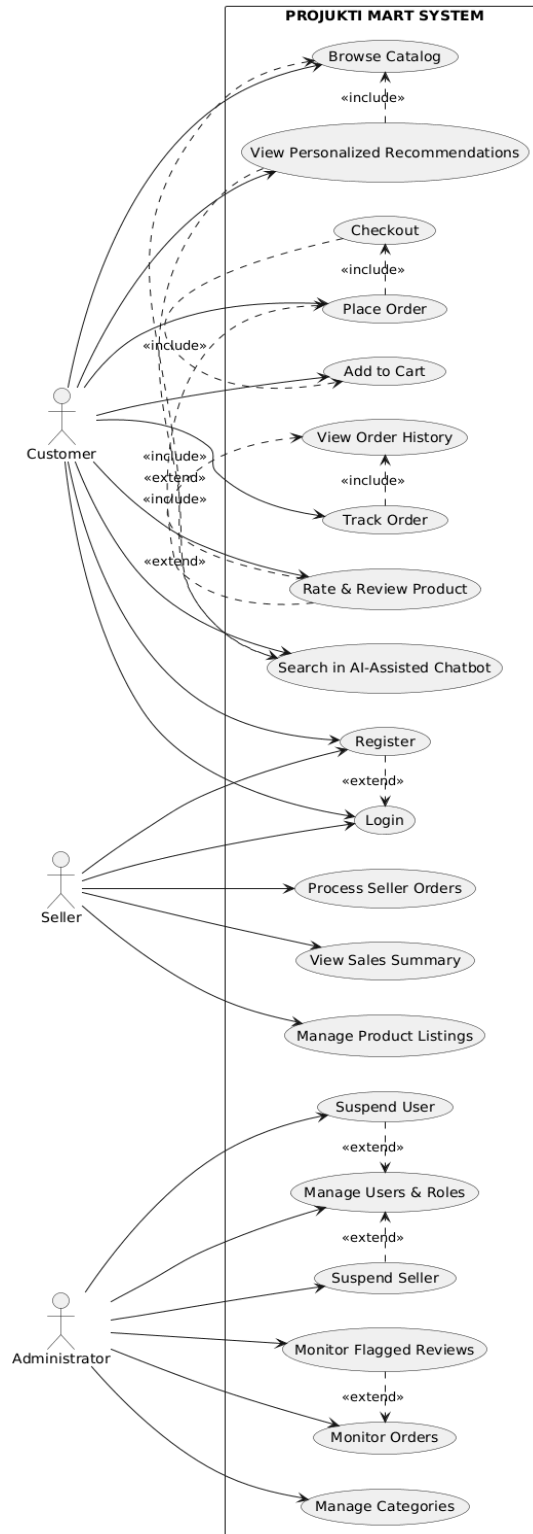
- **Input:** userId or sessionId, context productId (optional), max results.
- **Output:** list of productIds with optional reason codes.
- **Failure mode:** return empty list; UI simply hides the recommendation carousel.

4(a).3 || Hardware / Software Interfaces

The prototype runs on standard server hardware or a developer workstation. The web front-end targets modern browsers. If an external AI API is used, HTTPS access to that API is required. Persistent storage is a relational database.

4(b). Modeling & Design Diagrams

4(b).1 || Use Case Diagram



4(b).2 || Use Case Descriptions

UC-01: AI-Assisted Product Search

Field	Description
Actor	Customer (primary), Guest
Precondition	User is on any page that contains the search bar. The catalog contains products.
Trigger	User types a natural-language query (e.g., “lightweight laptop for programming under 80k”) and submits.
Main Flow	1. The system sends a query + optional context to the AI Semantic Search service. 2. AI returns a ranked product list with scores. 3. If confidence is acceptable, the system displays results labeled “AI Search results.” 4. Users may click a product to view details.
Alternate / Exception	• AI timeout or low confidence → system automatically runs keyword search and shows “Showing keyword results (AI unavailable)”. • Empty result set → display “No products matched your query” with suggestions to broaden filters.
Postcondition	Search results are displayed. Query is optionally logged for future personalization.

UC-02: Add to Cart & Checkout

Field	Description
Actor	Customer (must be logged in for checkout)
Precondition	The product exists and has stockQty > 0. Users may already have items in their cart.
Trigger	The user clicks “Add to Cart” on a product page or listing, then proceeds to Checkout.
Main Flow	1. System validates requested quantity ≤ available stock. 2. Items are added/updated in the cart. 3. The user navigates to the cart, reviews items, and updates quantities if desired. 4. User proceeds to checkout, confirms the shipping address, and orders a summary.

	5. System re-validates stock, creates Order + OrderItems, decrements stock, and clears cart. 6. The confirmation page is shown with the order ID.
Alternate / Exception	• Quantity exceeds stock → error message, quantity capped or rejected. • Stock changes between cart and checkout → system re-checks and notifies users of any items that became unavailable.
Postcondition	The order is recorded in the Pending/Confirmed state. Stock is reduced. The cart is empty.

UC-03: Seller Manages Product Listing

Field	Description
Actor	Seller
Precondition	The seller is authenticated, and the account is active.
Trigger	Seller navigates to “My Products” and chooses Create / Edit / Deactivate
Main Flow	1. Seller fills product form (title, description, brand, category, price, stock, key specs, images). 2. System validates required fields and price > 0, stock ≥ 0. 3. Product is saved and becomes visible in the catalog (if active). 4. Seller may later edit any field or set the status to Inactive.
Alternate / Exception	• Validation failure → form redisplay with error messages. • Attempt to edit another seller’s product → authorization error.
Postcondition	Product records are created or updated. The catalog reflects the change.

UC-04: Verify and Process an Order (Seller)

Field	Description
Actor	Seller
Precondition	An order exists that contains at least one product belonging to the seller.
Trigger	Seller opens the “Incoming Orders” dashboard and selects an order.
Main Flow	1. The system displays order details filtered to the seller’s line items. 2. Seller updates status (e.g., Confirmed → Shipped) and may add tracking notes. 3. The system records the status change and timestamp. 4. Customers can later see the updated status in the order history.
Alternate / Exception	• Seller attempts to change an order that contains none of their products → access denied.
Postcondition	Order status is updated. Audit trail is retained.

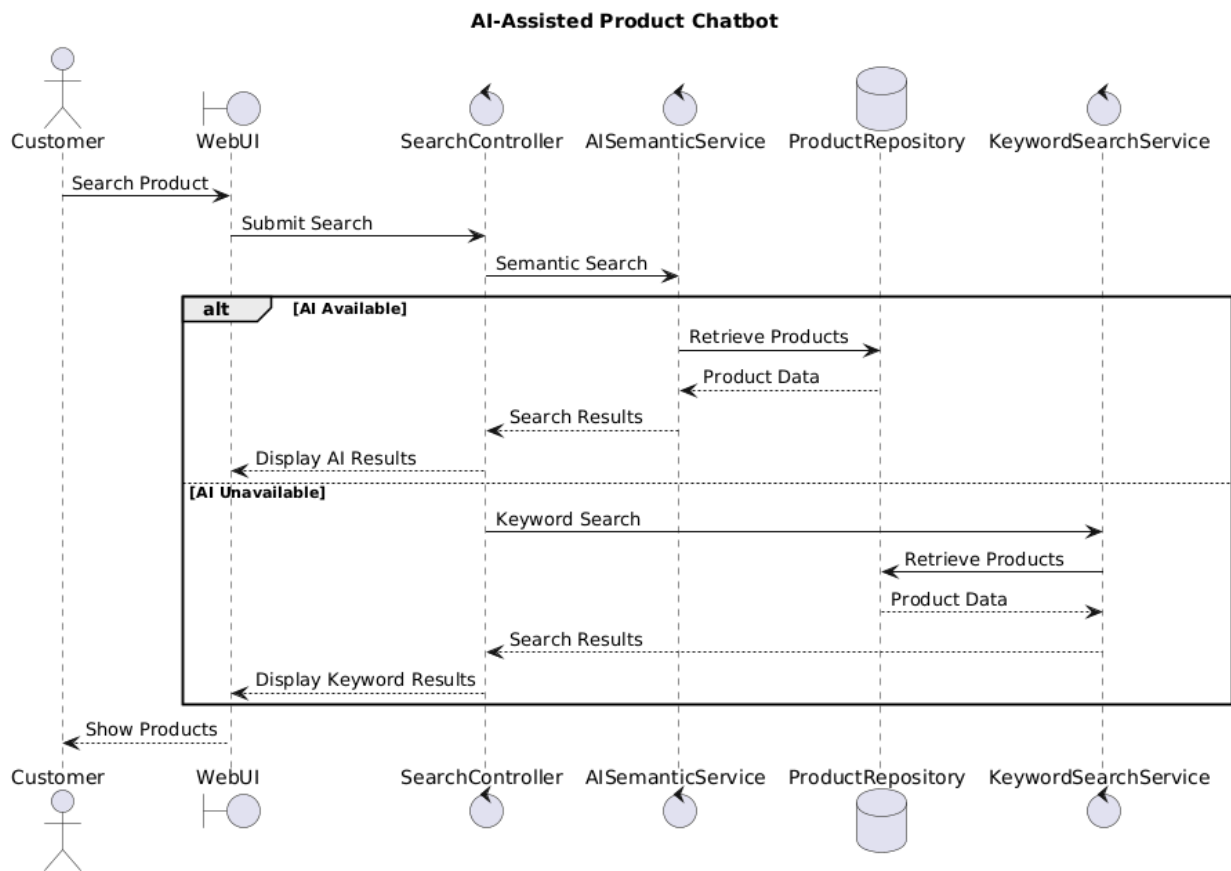
UC-05: View Personalized Recommendations

Field	Description
Actor	Customer (logged-in or anonymous session)
Precondition	The user has browsing or purchase history, or is viewing a product detail page.
Trigger	The user lands on the Home page or Product Detail page.
Main Flow	1. System requests recommendations from the Recommendation service (userId / session + optional context product). 2. Service returns a list of product IDs. 3. System renders a “Recommended for you” or “Customers also viewed” carousel. 4. Users may click any recommended product.
Alternate / Exception	• Service returns an empty list or is unavailable → carousel is hidden; no error is shown to the user.
Postcondition	Recommendations are displayed (or silently omitted).

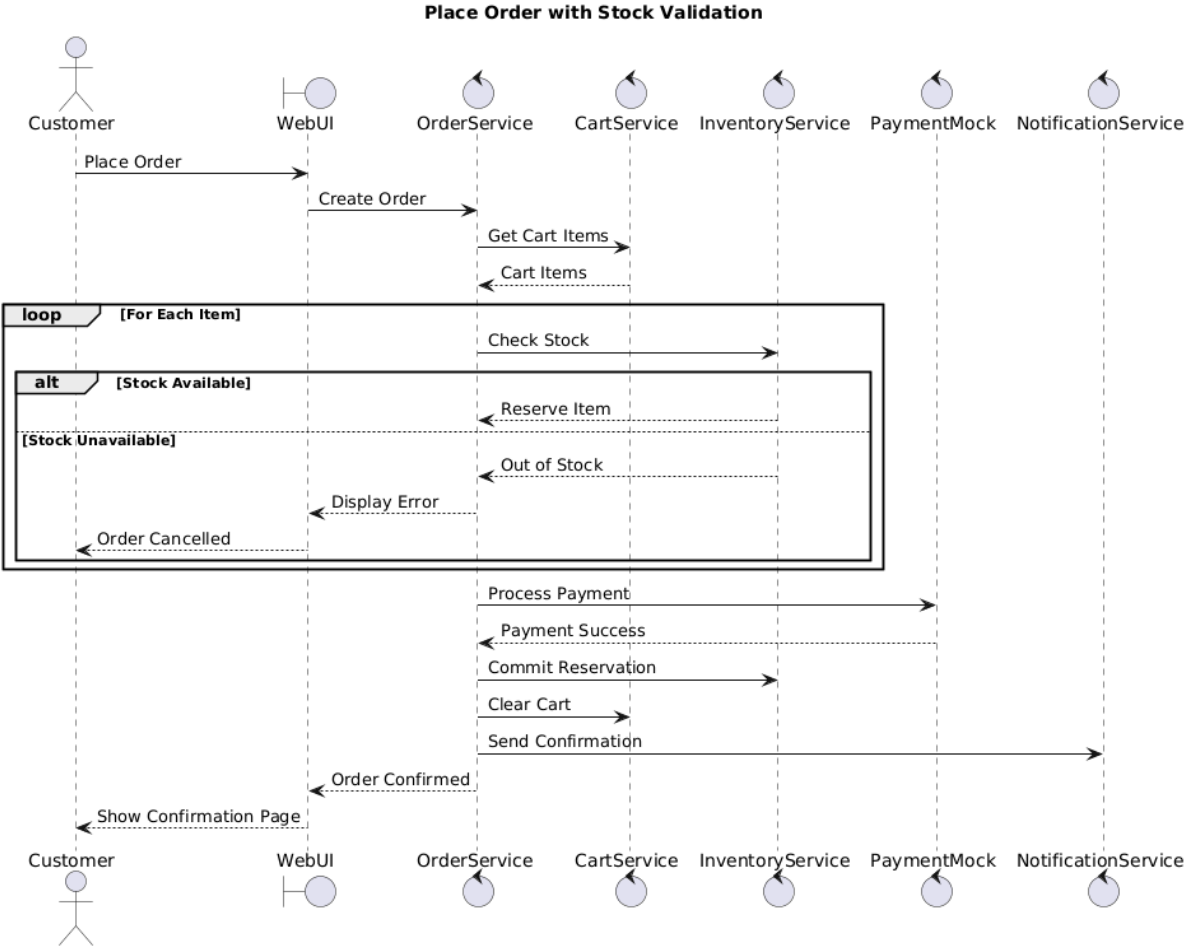
4(b).3 || Sequence Diagrams

Two major interactions are documented. The first concerns the AI component; the second covers the critical order-placement path, including stock validation.

Sequence Diagram 1: AI-Assisted Product Search

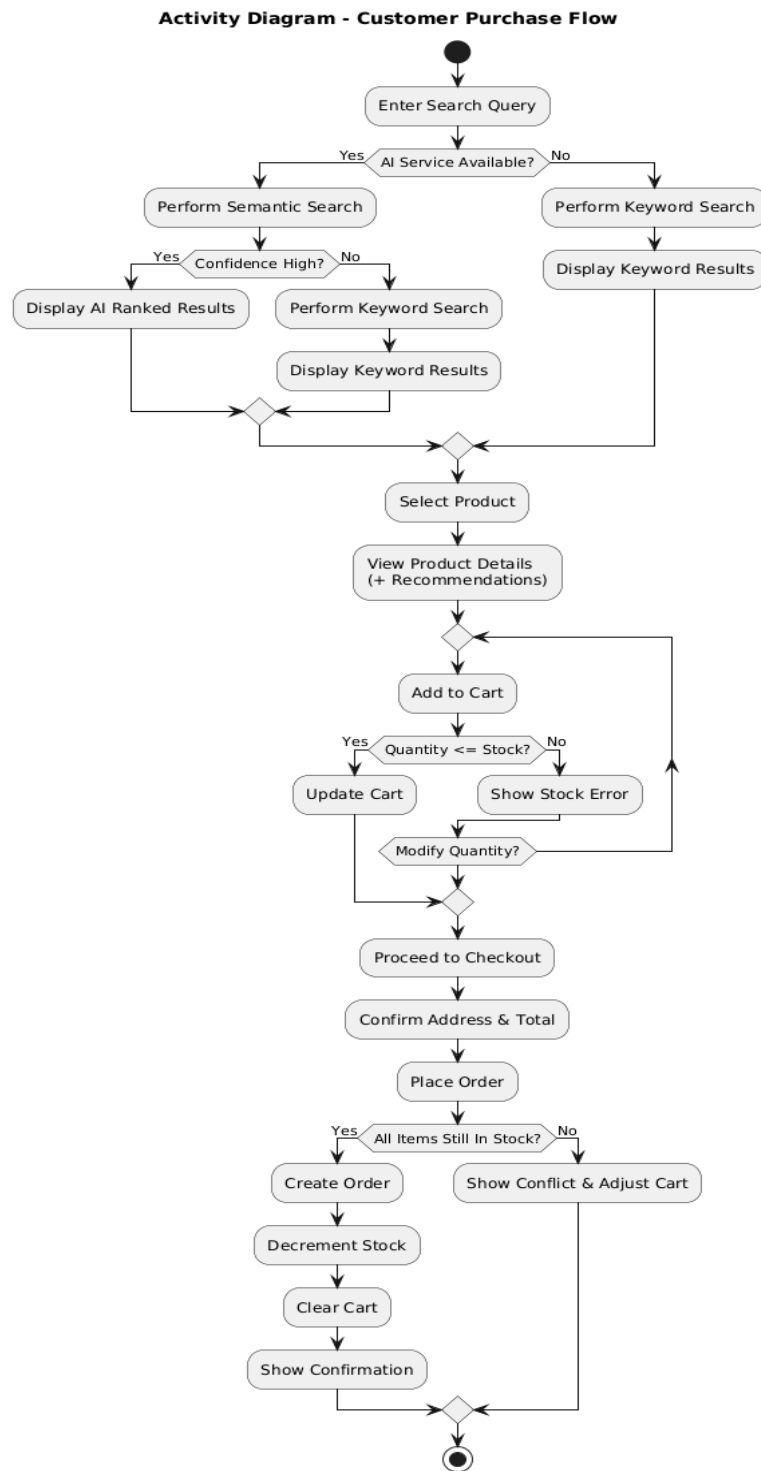


Sequence Diagram 2: Place Order (with stock validation)



4(b).4 || Activity Diagrams

Activity Diagram 1: Customer Purchase Flow (Search → Order)



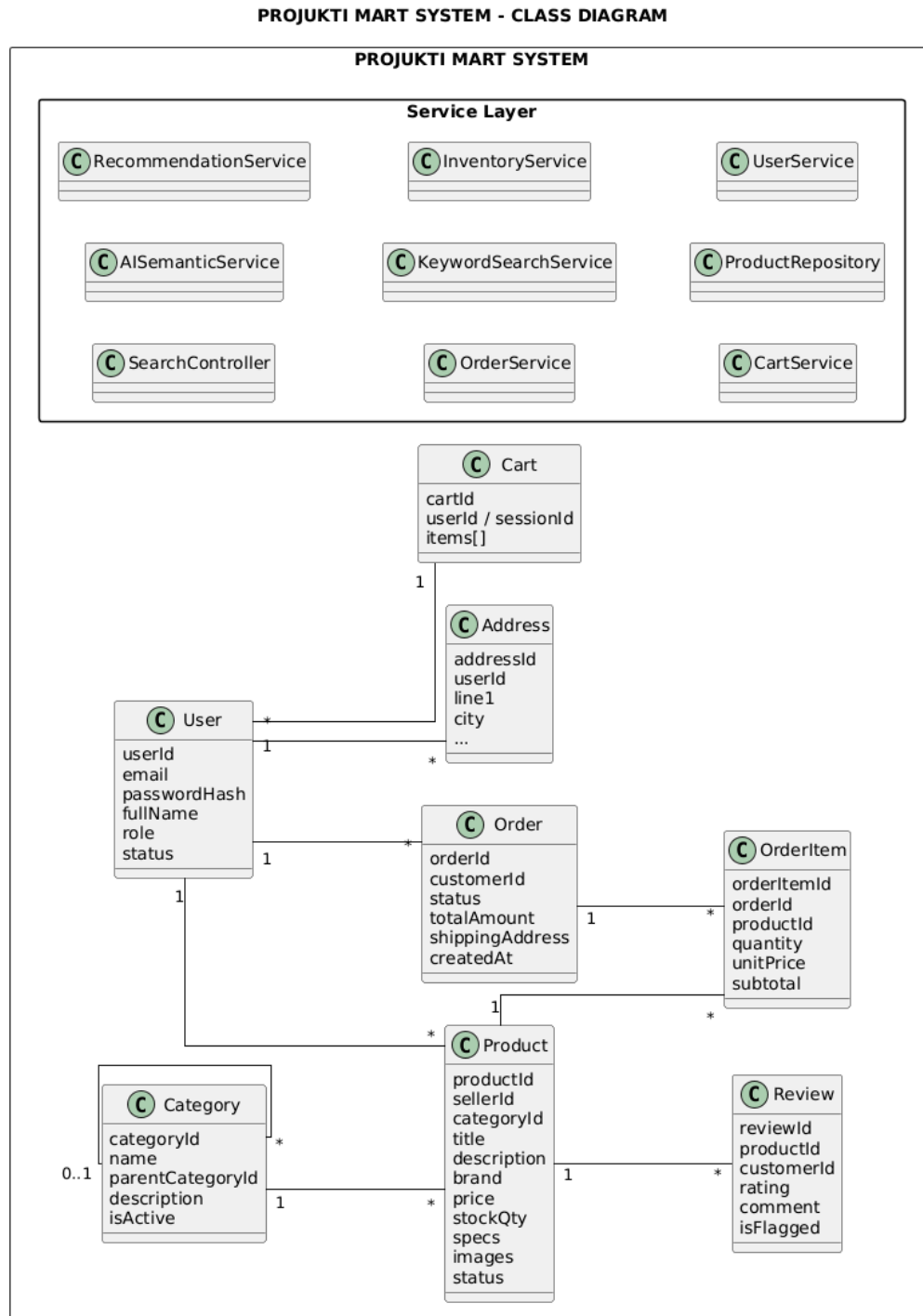
Activity Diagram 2: Seller Product Listing Lifecycle

Activity Diagram - Seller Product Listing Lifecycle



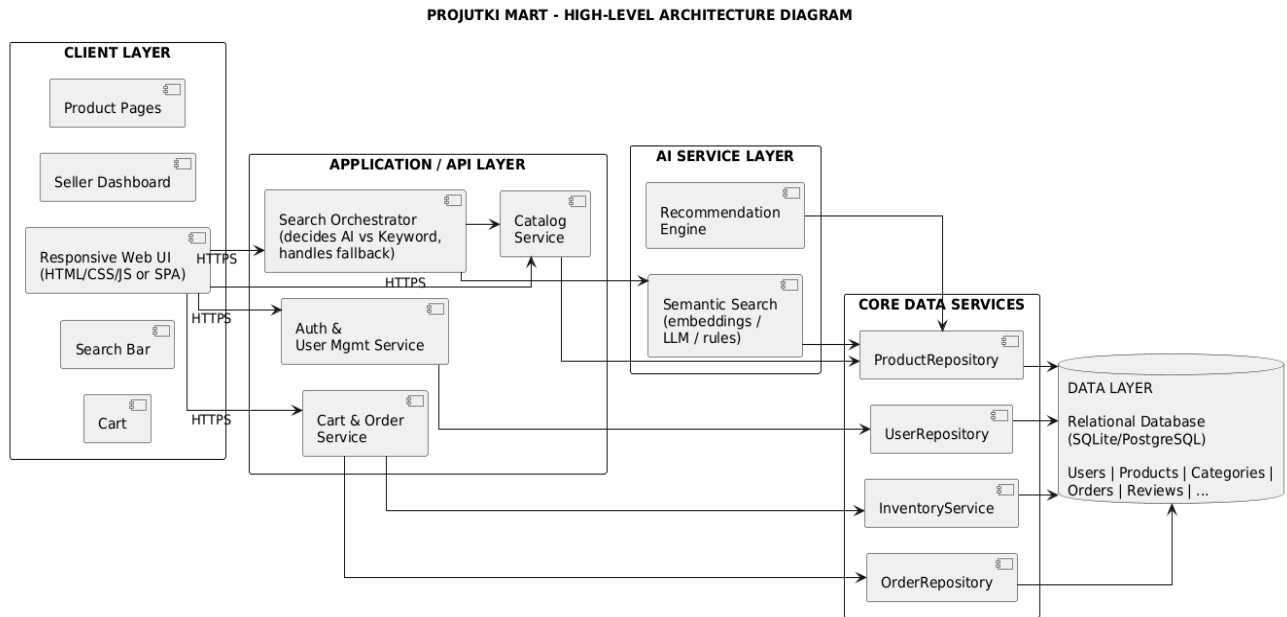
4(b).5 || Class Diagram

The principal domain and service classes are listed below, along with key attributes and relationships. In a formal UML tool, this would be drawn with association, aggregation, and inheritance connectors.



4(b).6 || High-Level Architecture Diagram

The architecture follows a layered approach, with a clearly isolated AI component to enable graceful degradation.



4(b).7 || UI Wireframes / Mockups (Conceptual)

Screen 1: Product Search / AI Assistant Interface

- Top header: Logo “**Projukti Mart**”, persistent search bar with placeholder “Describe what you need... (e.g., lightweight laptop for coding under 70k)”, cart icon, user menu.
- Below the search bar: quick category chips (Laptops, PCs, Keyboards, Headphones, Cables, Stands).
- Results area: left-side filter panel (price slider, brand, RAM, storage, etc.); main area shows product cards. A subtle badge “Keyword results” appears above the grid.
- Each card: image, title, 2–3 key specs, price, rating stars, “Add to Cart” button.

Screen 2: Product Listing / Detail Page with Recommendations

- Breadcrumb: Home > Laptops > Gaming Laptops > Product Title.
- Left/main: large image gallery, title, price, stock status, “Add to Cart” / quantity selector, key specification table, full description, customer reviews.
- Right or below: “Recommended for you” horizontal carousel of product cards driven by the recommendation service.
- “Customers who viewed this also viewed” secondary carousel.

Screen 3: Cart & Checkout Screen

- Cart page: table/list of items (image, title, unit price, quantity spinner, line subtotal, remove).
- Order summary panel: subtotal, shipping estimate, tax, total, “Proceed to Checkout” button.
- Checkout step: select/add shipping address, review final totals, “Place Order” (mock payment).
- Success page: order ID, estimated delivery text, link to order history.

Screen 4: Seller Product Management Dashboard

- Sidebar navigation: Dashboard, My Products, Incoming Orders, Sales Summary, Account.
- My Products view: table of seller’s products (image thumbnail, title, price, stock, status, Edit / Deactivate actions). “Add New Product” button opens a form with fields for all required attributes, including structured specs (brand, model, key technical attributes).
- Incoming Orders: list of orders containing the seller’s items with status update controls.
- Sales Summary: simple cards or table showing units sold and revenue for a selectable period.